LEARNING HUB MASTER REFERENCE SERIES • VOLUME 02

Linear Algebra: Solving Linear Equations & Elimination

Vectors and Linear Systems, The Row and Column Pictures, Gaussian Elimination & Pivots, Elementary Matrices, The 4 Perspectives of Matrix Multiplication, Invertibility & Gauss-Jordan, Triangular Factorizations (A = LU, A = LDU, PA = LU), and Flop Complexity Analysis

---

Primary Source Textbooks & Authorities:

- Gilbert Strang — *Introduction to Linear Algebra* (5th Edition, Chapter 2)
- Kenji Hiranabe — *The Art of Linear Algebra* (Visual Matrix Operations & The 4 Perspectives)
- MIT OpenCourseWare 18.06 — *Linear Algebra Lecture Series*

Standard Features:

- Pure Vector SVG Diagrams (Row vs. Col, Elimination Pipeline, Factorization Schematics, 4 Perspectives)
- 5 Hand-Calculation Fast-Track Shortcut Cards (⚡ TRICKS 1 – 5)
- 5 Rigorous, Multi-Step Worked Problems with Complete Tableaux & Sanity Checks

Table of Contents

Part I: Vectors, Linear Equations & The Two Geometric Pictures

Chapter 1: The Geometry of Linear Equations (Row vs. Column Picture)

Chapter 2: Solvability of A**x** = **b** & The Column Space C(A) ⚡ TRICK 1

Part II: The Idea and Algorithm of Elimination

Chapter 3: Gaussian Elimination, Pivots & Forward Multipliers

Chapter 4: Back-Substitution, Triangular Systems & Breakdown Analysis ⚡ TRICK 2

Part III: Matrix Operations, Elementary Matrices & The 4 Perspectives

Chapter 5: Elimination Matrices E_{ij}, Inverse Operations & Permutations P_{ij}

Chapter 6: The 4 Perspectives of Matrix Multiplication & Block Matrices ⚡ TRICK 3

Part IV: Inverses, Gauss-Jordan & Triangular Factorization

Chapter 7: Invertibility Criteria & The Gauss-Jordan Inversion Algorithm ⚡ TRICK 4

Chapter 8: The Great Factorization: A = LU and A = LDU ⚡ TRICK 5

Chapter 9: Transposes, Symmetric Matrices ($A = LDL^T$), and Row Exchanges (PA = LU)

Part V: Step-by-Step Worked Problems & Reference Appendices

Chapter 10: 5 Complete Worked Problems with Step-by-Step Tableaux

Appendix A: Computational Complexity Analysis: Flop Counting (~⅓ n^3 vs n^2)

Appendix B: Master Hand-Calculation Fast-Track Cheat Sheet & Matrix Rules

Part I: Vectors, Linear Equations & The Two Geometric Pictures

Chapter 1: The Geometry of Linear Equations (Row vs. Column Picture)

The central problem of linear algebra is to solve a system of n linear equations in n unknowns:

$$\begin{aligned} a_{11}\,x_1 + a_{12}\,x_2 + \dots + a_{1n}\,x_n &= b_1 \\ a_{21}\,x_1 + a_{22}\,x_2 + \dots + a_{2n}\,x_n &= b_2 \\ &\vdots \\ a_{n1}\,x_1 + a_{n2}\,x_2 + \dots + a_{nn}\,x_n &= b_n \end{aligned}$$

In matrix notation, this entire system collapses into the single, universally celebrated equation:

$$\mathbf{A\,x = b} \quad \Leftrightarrow \quad \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

To truly master linear algebra, one must see this equation from two distinct geometric viewpoints: the **Row Picture** and the **Column Picture**. As Professor Gilbert Strang emphasizes, the row picture is familiar from high school geometry, but the column picture is infinitely more fruitful for understanding solvability, vector spaces, and modern computation.

1.1 The Row Picture: Intersecting Hyperplanes

In the Row Picture, each equation is viewed independently as an $(n-1)$-dimensional geometric surface (a hyperplane) living in n-dimensional space $\mathbb{R}^n$:

- **In 2D ($n = 2$):** Each linear equation $a\,x + b\,y = c$ defines a 1D straight line. Two equations define two lines. The unique solution (x^*, y^*) is the single geometric point where the two lines cross. If the lines are parallel, they never meet (no solution); if they are identical, they intersect along the whole line (infinitely many solutions).
- **In 3D ($n = 3$):** Each linear equation $a\,x + b\,y + c\,z = d$ defines a flat 2D plane. The first two planes typically intersect along a straight line L. The third plane intersects that line L at a single point (x^*, y^*, z^*).
- **In n dimensions:** Each equation represents an $(n-1)$-dimensional hyperplane perpendicular to the row normal vector $\mathbf{r}_i = [a_{i1}, \dots, a_{in}]$. The solution $\mathbf{x}$ is the intersection of all n hyperplanes: $\bigcap_{i=1}^{n} \{\, \mathbf{x} \in \mathbb{R}^n \mid \mathbf{r}_i \cdot \mathbf{x} = b_i \,\}$.

1.2 The Column Picture: Linear Combinations of Column Vectors

In the Column Picture, we rewrite $A\mathbf{x} = \mathbf{b}$ as a single vector equation:

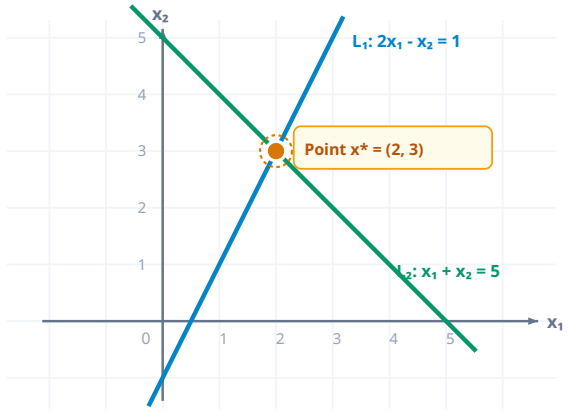
$$x_1 \begin{vmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{n1} \end{vmatrix} + x_2 \begin{vmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{n2} \end{vmatrix} + \ldots + x_n \begin{vmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{nn} \end{vmatrix} = \begin{vmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{vmatrix} \quad \Leftrightarrow \quad x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \ldots + x_n \mathbf{a}_n = \mathbf{b}$$

This shifts our perspective completely:

Instead of asking *"Where do n hyperplanes intersect?"*, we ask: **"How can we stretch and combine the column vectors $a_1, a_2, \ldots, a_n$ by scalar weights $x_1, x_2, \ldots, x_n$ so that their vector sum lands exactly on the target vector b?"**

ROW PICTURE: Intersecting Lines / Planes

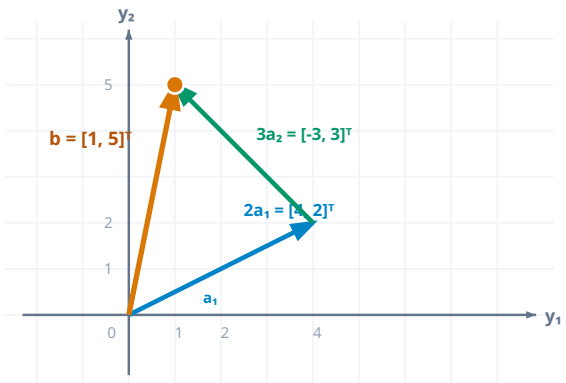
System: $2x_1 - x_2 = 1$ (L_1) and $x_1 + x_2 = 5$ (L_2)



Geometric Meaning: Each row is an (n-1)-dim hyperplane.
The unique solution is the single intersection point of all planes.

COLUMN PICTURE: Vector Combinations

Combination: $x_1 [2, 1]^T + x_2 [-1, 1]^T = [1, 5]^T = b$



Geometric Meaning: Find multipliers x_1, x_2 combining the columns.
A solution exists iff target vector b lies in Column Space C(A)!

Figure 1: The Dual Geometric Pictures of $2x_1 - x_2 = 1$ and $x_1 + x_2 = 5$. **Left:** The Row Picture shows intersecting lines meeting at (2, 3). **Right:** The Column Picture shows the linear combination $2\mathbf{a}_1 + 3\mathbf{a}_2$ assembling to hit target $\mathbf{b} = [1, 5]^T$.

◎ Geometric Supremacy of the Column Picture

While drawing hyperplanes in 4D or 100D is completely impossible for the human brain to visualize, the **Column Picture remains visual in any dimension**: we always have *n* column vectors living in the same space $\mathbb{R}^m$, and we are simply searching for the scalar coefficients $x_1, \ldots, x_n$ that combine them into **b**. All fundamental concepts of linear algebra (linear independence, basis, dimension, span, column space, and rank) arise directly from the Column Picture.

Chapter 2: Solvability of Ax = b & The Column Space C(A)

The linear combination formulation gives an immediate, definitive answer to the question of solvability:

▶ Fundamental Solvability Condition (Strang's Theorem 2.1)

The linear system **A x = b** has a solution if and only if the target vector **b** can be expressed as a linear combination of the columns of **A**. In modern terms:

$$\mathbf{Ax = b \text{ is solvable}} \quad \Leftrightarrow \quad \mathbf{b} \in C(A)$$

where $C(A) \subseteq \mathbb{R}^m$ is the **Column Space of A** (the subspace spanned by all columns of A).

For a square $n \times n$ matrix A, there are two mutually exclusive possibilities:

Attribute	Non-Singular (Invertible) Case	Singular (Non-Invertible) Case
Column Independence	Columns $\mathbf{a}_1, \ldots, \mathbf{a}_n$ are linearly independent.	Columns are linearly dependent (at least one column is a combination of others).
Column Space C(A)	Spans the entire space: $C(A) = \mathbb{R}^n$.	Spans a restricted subspace of dimension $r < n$ (a line or plane).
Solvability for b	**Always solvable** for *any* right-hand side $\mathbf{b} \in \mathbb{R}^n$.	**Solvable only if b** accidentally falls inside the subspace $C(A)$.
Number of Solutions	**Unique solution:** $\mathbf{x}^* = A^{-1}\mathbf{b}$.	Either **zero solutions** (inconsistent) or **infinitely many solutions**.
Determinant & Pivots	$\det(A) \neq 0$; exactly n non-zero pivots exist.	$\det(A) = 0$; fewer than n pivots survive elimination.

TRICK 1 **The 2×2 Determinant & Invertibility Test** — **Instant Δ = ad - bc check**

Standard Method (Slow): Setting up augmented row reduction or computing minors to test whether a 2×2 system has a unique solution.

⚡ **Fast Hand Calculation Shortcut:** For any 2×2 matrix $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$, immediately compute the determinant $\Delta = ad - bc$. If **ad − bc ≠ 0**, the columns are independent, lines intersect at a single point, and a unique solution is 100% guaranteed. If **ad − bc = 0**, the rows/columns are parallel: test if {frac('a', 'c')} = {frac('b', 'd')} = {frac('b_1', 'b_2')}. If the right-hand sides match the ratio → Infinitely many solutions. If the ratio fails → Zero solutions (parallel lines).

Part II: The Idea and Algorithm of Elimination

Chapter 3: Gaussian Elimination, Pivots & Forward Multipliers

Gaussian Elimination is the systematic algorithm used by humans and supercomputers alike to solve $A\mathbf{x} = \mathbf{b}$. Instead of tackling coupled, tangled equations, elimination applies elementary row operations that preserve the solution set while converting the coefficient matrix A into an **upper triangular matrix U**.

3.1 The Three Elementary Row Operations

The algebraic validity of Gaussian elimination rests on the fact that three operations leave the solution set of a linear system unchanged:

1. **Row Replacement:** Add a scalar multiple of row j to row i: $R_i \leftarrow R_i - l_{ij} R_j$.
2. **Row Exchange (Permutation):** Swap two rows: $R_i \leftrightarrow R_j$ (required when a pivot position contains zero).
3. **Row Scaling:** Multiply all entries in row i by a non-zero scalar: $R_i \leftarrow c\, R_i$ ($c \neq 0$).

3.2 Pivots and Multipliers

Elimination proceeds column by column from left to right:

- **Pivot 1:** The entry in the top-left corner, $d_1 = a_{11}$, is the first pivot. It must be non-zero (if zero, exchange rows).
- **Column 1 Elimination:** To eliminate the entry a_{i1} below the pivot in row i, we define the **multiplier**:

$$l_{i1} = \frac{\text{entry to eliminate}}{\text{pivot 1}} = \frac{a_{i1}}{a_{11}}$$

  Subtracting l_{i1} times row 1 from row i creates a zero in position $(i, 1)$:

$$\text{Row}_i \leftarrow \text{Row}_i - l_{i1}\, \text{Row}_1$$

- **Pivot 2:** The new entry in position (2, 2) becomes the second pivot $d_2 = a_{22}^{(1)}$.
- **Column 2 Elimination:** We compute multipliers $l_{i2} = a_{i2}^{(1)} / d_2$ for all rows $i > 2$ and subtract.
- **General Step k:** With pivot $d_k = a_{kk}^{(k-1)}$, multipliers are:

$$l_{ij} = \frac{a_{ij}^{(k-1)}}{a_{jj}^{(k-1)}} \qquad \text{\text{for all } i > k}$$

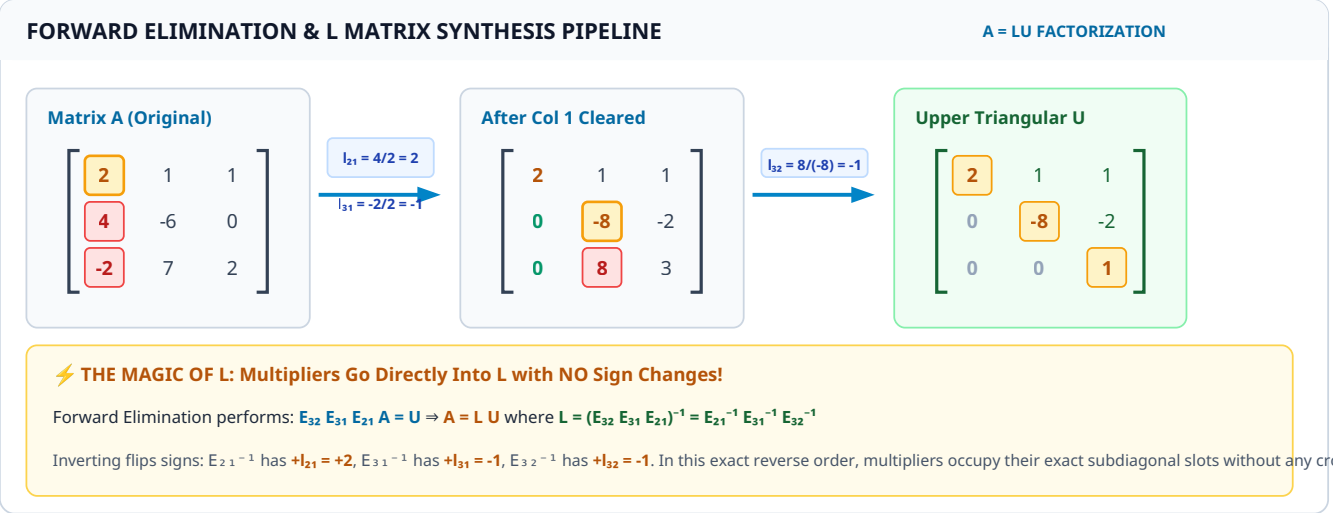


Figure 2: Forward Elimination Pipeline. Pivots (highlighted in gold) systematically zero out entries below them (red) via multipliers l_{ij}, resulting in upper triangular matrix U. The multipliers drop directly into L.

Chapter 4: Back-Substitution, Triangular Systems & Breakdown Analysis

4.1 Back-Substitution: Solving Ux = c

Once forward elimination is complete, the augmented system $[A \mid \mathbf{b}]$ has been transformed into the upper triangular system $[U \mid \mathbf{c}]$:

$$\begin{vmatrix} u_{11} & u_{12} & \dots & u_{1n} \\ 0 & u_{22} & \dots & u_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & u_{nn} \end{vmatrix} \begin{vmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{vmatrix} = \begin{vmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{vmatrix}$$

Because the last equation contains only one unknown $u_{nn}\, x_n = c_n$, we solve backwards:

$$x_n = \frac{c_n}{u_{nn}}, \quad x_{n-1} = \frac{c_{n-1} - u_{n-1,n}\, x_n}{u_{n-1,n-1}}, \quad \dots \quad x_i = \frac{c_i - \sum_{j=i+1}^{n} u_{ij}\, x_j}{u_{ii}}$$

Back-substitution is remarkably fast: it requires only $\sim n^2/2$ multiplications and subtractions, compared to the $\sim n^3/3$ operations required by forward elimination.

4.2 Breakdown of Elimination: Temporary vs. Permanent

What happens when a zero appears in the pivot position ($a_{kk} = 0$)? We must distinguish between two fundamentally different situations:

Breakdown Type	Mathematical Mechanism	Diagnostic & Resolution
Temporary Breakdown (Non-Singular)	A zero appears in pivot position (k, k), but there is a **non-zero entry below it** in row $p > k$.	**Cured by Row Exchange:** Swap row k with row p (multiplication by permutation matrix P_{kp}). Elimination resumes normally. A is invertible!
Permanent Breakdown (Singular)	A zero appears in pivot position (k, k), and **all entries below it in column k are also zero**.	**Fatal: No non-zero pivot can be found.** Matrix A is singular (det A = 0). If the right-hand side in the zero row is 0 → ∞ solutions. If the right-hand side is non-zero → 0 solutions (inconsistent).

⚠ Numerical Pitfall: Small Pivots and Partial Pivoting

On a computer using floating-point arithmetic, a pivot that is *very small but not exactly zero* (e.g. 10^{-15}) is just as dangerous as an exact zero! Dividing by a tiny pivot produces huge multipliers ($l_{ij} \sim 10^{15}$), causing catastrophic round-off error that obliterates the true solution.

Remedy (Partial Pivoting): In column k, search all rows from k to n for the entry with the **largest absolute value** $|a_{ik}|$, and swap that row into the pivot position. This guarantees $|l_{ik}| \leq 1$, ensuring rock-solid numerical stability.

TRICK 2 Mental Check: Count the Non-Zero Pivots

det(A) = product of pivots

Standard Method (Slow): Expanding a 3×3 or 4×4 determinant with minors to see if the matrix is invertible.

⚡ **Fast Hand Calculation Shortcut:** Count the surviving pivots during forward elimination! If you successfully obtain **n non-zero pivots** ($d_1, d_2, \ldots, d_n \neq 0$), the matrix is guaranteed invertible, the determinant is simply the product of the pivots **det(A) = (±1) $d_1 d_2 \ldots d_n$** (sign depends on number of row exchanges), and the nullspace contains only the zero vector **x = 0**.

Part III: Matrix Operations, Elementary Matrices & The 4 Perspectives

Chapter 5: Elimination Matrices E_{ij}, Inverse Operations & Permutations P_{ij}

The true elegance of matrix algebra is that **row operations are performed by matrix multiplication on the left**.

5.1 The Elementary Matrix E_{ij}

The elementary matrix E_{ij} is constructed by starting with the Identity matrix I and inserting $-l_{ij}$ into row i, column j:

$$E_{21} = \begin{vmatrix} 1 & 0 & 0 \\ -l_{21} & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} \quad \&implies; \quad E_{21}\,A = \begin{vmatrix} \text{Row 1 of A} \\ \text{Row 2} - l_{21}\,\text{Row 1} \\ \text{Row 3 of A} \end{vmatrix}$$

5.2 The Inverse of an Elementary Matrix: E_{ij}^{-1}

To undo the operation *"subtract l_{ij} times row j from row i"*, the inverse operation is simply: **"add l_{ij} times row j back to row i"**. Therefore, to invert an elementary matrix, **we simply flip the sign of the off-diagonal entry**:

$$E_{21} = \begin{vmatrix} 1 & 0 & 0 \\ -l_{21} & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} \quad \&implies; \quad E_{21}\^\{-1\} = \begin{vmatrix} 1 & 0 & 0 \\ +l_{21} & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

5.3 Permutation Matrices P_{ij}

A permutation matrix P exchanges rows when multiplying on the left, and exchanges columns when multiplying on the right. It is created by swapping rows of the identity matrix I:

$$P_{23} = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{vmatrix} \quad \&implies; \quad P_{23}\,A = \begin{vmatrix} \text{Row 1 of A} \\ \text{Row 3 of A} \\ \text{Row 2 of A} \end{vmatrix}$$

▶ **Key Properties of Permutation Matrices**

For any permutation matrix P:

1. Every row and every column contains exactly one 1 and all other entries 0.
2. The inverse of a permutation matrix is its transpose: $\mathbf{P^{-1} = P^T}$.
3. There are $n!$ distinct $n \times n$ permutation matrices, corresponding to all possible permutations of n objects.

Chapter 6: The 4 Perspectives of Matrix Multiplication & Block Matrices

Matrix multiplication $AB = C$ is often introduced mechanically as "row times column". However, in modern linear algebra and data science, **there are four distinct, deeply illuminating perspectives** to compute and understand AB:

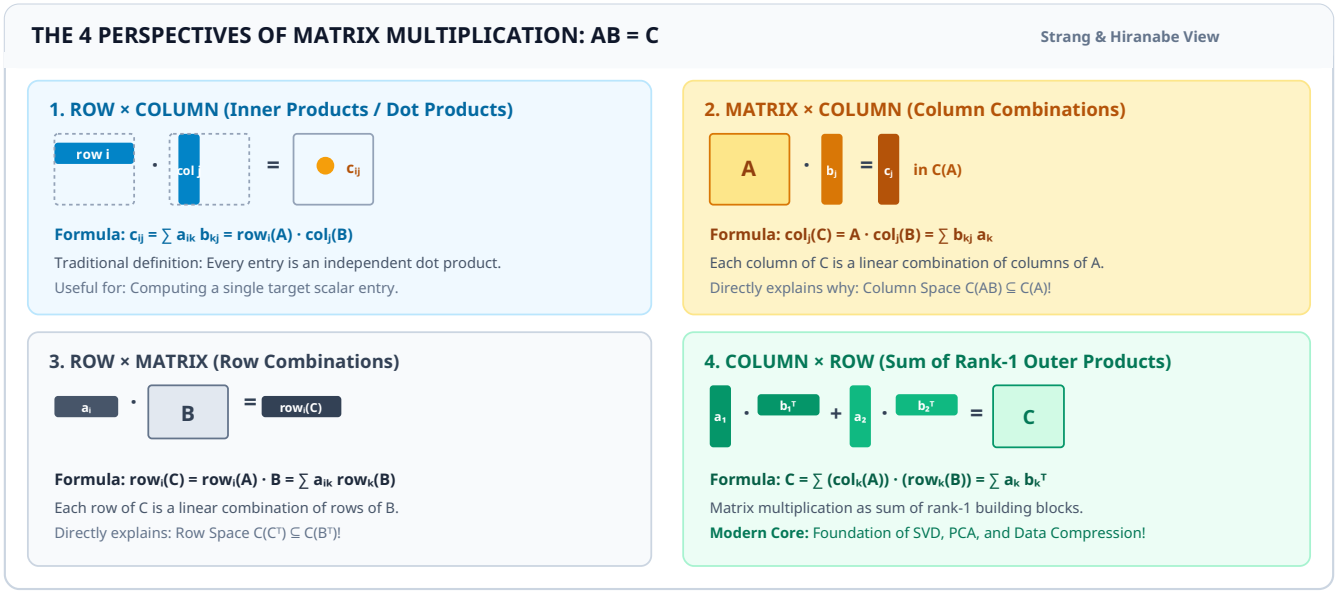


Figure 3: The 4 Perspectives of Matrix Multiplication $AB = C$ (Strang & Hiranabe). Each perspective illuminates a distinct algebraic structure: inner products, column spaces, row spaces, and rank-1 outer product sums.

Perspective 1: Row × Column (Inner Product / Entry-by-Entry)

Each entry c_{ij} is the dot product of row i of A with column j of B:

$$c_{ij} = \text{Row}_i(A) \cdot \text{Col}_j(B) = \sum_{k=1}^p a_{ij}\, b_{jj}$$

Use case: Ideal when you only need to compute a single specific entry in the product matrix without computing the rest.

Perspective 2: Matrix × Column (Column Combinations)

The product matrix C is computed column-by-column: each column of C is the matrix A multiplying a column of B:

$$C = A\,[\,\mathbf{b}_1 \;\; \mathbf{b}_2 \;\; \dots \;\; \mathbf{b}_n\,] = [\,A\mathbf{b}_1 \quad A\mathbf{b}_2 \quad \dots \quad A\mathbf{b}_n\,]$$

Since $A\mathbf{b}_j$ is a linear combination of the columns of A, this immediately proves the vital theorem:

> ▶ **Column Space Inclusion**
>
> Every column of AB is a linear combination of the columns of A. Therefore: **$\mathbf{C(AB) \subseteq C(A)}$**.

Perspective 3: Row × Matrix (Row Combinations)

The product matrix C is computed row-by-row: each row of C is a row of A multiplying the entire matrix B:

$$C = \begin{vmatrix} \text{Row}_1(A) \\ \text{Row}_2(A) \\ \vdots \end{vmatrix} \quad B = \begin{vmatrix} \text{Row}_1(A)\, B \\ \text{Row}_2(A)\, B \\ \vdots \end{vmatrix}$$

Each row of C is a linear combination of the rows of B, proving: $\mathbf{C((AB)^T) \subseteq C(B^T)}$.

Perspective 4: Column × Row (Sum of Rank-1 Outer Products)

This is Kenji Hiranabe and Gilbert Strang's most profound perspective: Multiply column k of A (an $m \times 1$ vector) by row k of B (a $1 \times n$ vector). Their product is an $m \times n$ matrix of **rank 1**. Summing all p outer products produces the full matrix C:

$$AB = \sum_{k=1}^p (\text{Col}_k(A)) (\text{Row}_k(B)) = \mathbf{a}_1 \mathbf{b}_1^T + \mathbf{a}_2 \mathbf{b}_2^T + \dots + \mathbf{a}_p \mathbf{b}_p^T$$

TRICK 3 **Rank-1 Outer Product Mental Decomposition** — Instant sum of rank-1 matrices

Standard Method (Slow): Carrying out 8 to 27 separate row-by-column dot products across whole matrices.

⚡ **Fast Hand Calculation Shortcut:** For matrices with sparse or clean columns, sum the outer products column-by-row! Example: $\begin{vmatrix} 1 & 0 \\ 2 & 3 \end{vmatrix} \begin{vmatrix} 4 & 5 \\ 0 & 1 \end{vmatrix} = \begin{vmatrix} 1 \\ 2 \end{vmatrix} \begin{vmatrix} 4 & 5 \end{vmatrix} + \begin{vmatrix} 0 \\ 3 \end{vmatrix} \begin{vmatrix} 0 & 1 \end{vmatrix} = \begin{vmatrix} 4 & 5 \\ 8 & 10 \end{vmatrix} + \begin{vmatrix} 0 & 0 \\ 0 & 3 \end{vmatrix} = \begin{vmatrix} 4 & 5 \\ 8 & 13 \end{vmatrix}$. Outer products are trivially written down by scaling the row by each element of the column!

6.5 Block Matrices & Block Multiplication

When matrices are partitioned into submatrix blocks, they multiply according to the exact same algebraic rules, provided the block dimensions match:

$$\begin{vmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{vmatrix} \begin{vmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{vmatrix} = \begin{vmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{vmatrix}$$

A celebrated application of block elimination is the **Schur Complement**: eliminating block A_{21} by subtracting $A_{21}A_{11}^{-1}$ times the first block row yields the reduced block $\mathbf{S = A_{22} - A_{21}A_{11}^{-1}A_{12}}$.

Part IV: Inverses, Gauss-Jordan & Triangular Factorization

Chapter 7: Invertibility Criteria & The Gauss-Jordan Inversion Algorithm

A square matrix A is **invertible** (or non-singular) if there exists a matrix A^{-1} such that:

$$A A^{-1} = I \quad \text{and} \quad A^{-1} A = I$$

7.1 Six Equivalent Invertibility Criteria

▶ The Master Invertibility Theorem for Square Matrices

For an $n \times n$ matrix A, the following conditions are 100% equivalent:

1. A has an inverse A^{-1}.
2. Gaussian elimination yields **n non-zero pivots** (no zero on the diagonal of U).
3. The determinant of A is non-zero: **det(A) ≠ 0**.
4. The only solution to $A\mathbf{x} = \mathbf{0}$ is the trivial solution **x = 0** (Nullspace $N(A) = \{\mathbf{0}\}$).
5. The columns of A are **linearly independent** (and span all of $\mathbb{R}^n$).
6. For every right-hand side **b**, the system $A\mathbf{x} = \mathbf{b}$ has a unique solution $\mathbf{x}^* = A^{-1}\mathbf{b}$.

7.2 Inverse of a Product: The Shoe-Sock Theorem

The inverse of a product reverses the multiplication order, exactly like taking off your shoes and socks:

$$(AB)^{-1} = B^{-1} A^{-1}$$

◆ Proof of Shoe-Sock Inversion

Multiply (AB) by $(B^{-1} A^{-1})$:

$$(AB)(B^{-1} A^{-1}) = A\,(B\,B^{-1})\,A^{-1} = A\,I\,A^{-1} = A\,A^{-1} = \mathbf{I}.$$

By uniqueness of the inverse, $(AB)^{-1} = B^{-1} A^{-1}$.

7.3 The Gauss-Jordan Inversion Algorithm

Gauss-Jordan computes A^{-1} by solving $A\mathbf{x}_j = \mathbf{e}_j$ for all n standard basis vectors simultaneously. We form the doubly-wide **augmented matrix [A | I]** and apply elementary row operations until A becomes I:

$$[\, A \mid I \,] \xrightarrow{\text{Forward Elimination}} [\, U \mid E \,] \xrightarrow{\text{Jordan Upward Elimination + Normalization}} [\, I \mid A^{-1} \,]$$

TRICK 4 **Gauss-Jordan Inversion Ledger Strategy** **Divide by pivots at end**

Standard Method (Slow): Writing out full equations or inverting via the cumbersome matrix of cofactors (adjugate).

⚡ **Fast Hand Calculation Shortcut:** In Gauss-Jordan, row operations applied to the left matrix A must be identically executed on the right identity matrix I. **Visual Ledger Shortcut:** Clear column 1 entirely (below AND above if doing Gauss-Jordan, or first reduce to U then clear upwards). Divide each row by its pivot at the very end to prevent messy fraction propagation during the elimination phase!

Chapter 8: The Great Factorization: A = LU and A = LDU

Elimination transformed A into U by applying elementary matrices on the left:

$$(E_{k} \dots E_{2} E_{1}) A = U$$

Moving all the elimination matrices to the other side gives:

$$A = (E_1^{-1} E_2^{-1} \dots E_k^{-1}) U = L U$$

▶ The Magic of the L Matrix (Strang's Elimination Miracle)

When forward elimination proceeds without row exchanges, the matrix:

$$L = E_{21}^{-1} E_{31}^{-1} \dots E_{n,n-1}^{-1}$$

is a **unit lower triangular matrix** with 1s on its main diagonal, and every multiplier l_{ij} sits **directly in entry (i, j) with NO sign reversal and NO cross-talk**!

Compare the product of E's with the product of E^{-1}'s:

$$E = E_{32} E_{31} E_{21} = \begin{vmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & -3 & 1 \end{vmatrix} \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{vmatrix} \begin{vmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 2 & -3 & 1 \end{vmatrix} \quad (\text{Messy mix!})$$

$$L = E_{21}^{-1} E_{31}^{-1} E_{32}^{-1} = \begin{vmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 4 & 0 & 1 \end{vmatrix} \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 3 & 1 \end{vmatrix} \quad (\text{Pure multipliers!})$$

In the inverse product L, each multiplier l_{ij} appears completely uncorrupted in position (i, j). **You never have to perform matrix multiplication or inversion to find L: you simply record the multipliers as you eliminate!**

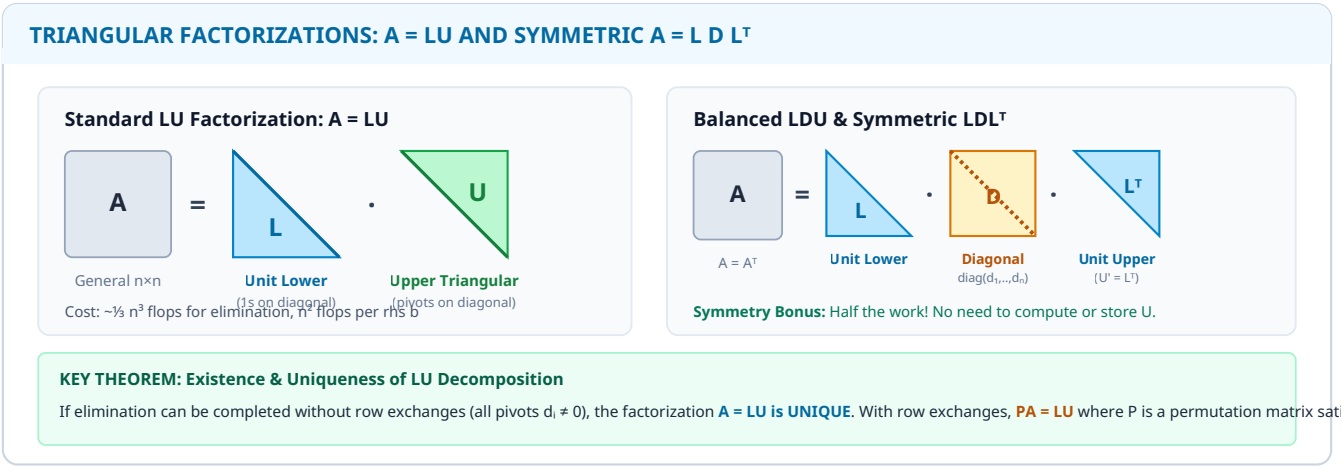


Figure 4: Triangular Factorizations $A = LU$ and $A = LDU$. Factoring out the pivot diagonal D produces balanced triangular factors with 1s on both diagonals. For symmetric matrices, $U' = L^T$, collapsing the factorization into $A = LDL^T$.

8.2 Solving Ax = b via LU: Two Triangular Systems

Once $A = LU$ is computed, solving $A\mathbf{x} = \mathbf{b}$ splits into two blazing-fast triangular solves:

1. **Forward Solve (L c = b):** Solve for intermediate vector **c** using forward substitution ($\sim n^2/2$ flops).
2. **Back Solve (U x = c):** Solve for final unknown vector **x** using back-substitution ($\sim n^2/2$ flops).

The Computational Payoff: If you need to solve $A\mathbf{x} = \mathbf{b}$ for 1,000 different vectors **b** (as in engineering simulations and control loops), you do **NOT** redo elimination $\sim 1{,}000 \times \frac{1}{3} n^3$ times! You compute $A = LU$ **once** (cost: $\frac{1}{3} n^3$), and then each subsequent solve costs only n^2 operations.

TRICK 5 **Instant 3×3 LU Assembly by Inspection** — **Drop multipliers directly into L**

Standard Method (Slow): Writing out separate E21, E31, E32 matrices and multiplying them explicitly.

⚡ **Fast Hand Calculation Shortcut:** As you eliminate entries below the pivots: 1. Write 1s on the diagonal of L. 2. Whenever you calculate multiplier $l_{ij} = a_{ij}/d_j$, immediately drop $+\mathbf{l_{ij}}$ into entry (i, j) of L! 3. Once elimination finishes, your final upper triangular matrix IS U. You are done! Check: (Row 1 of L) × (Col 1 of U) = $1 \cdot u_{11} = a_{11}$.

Chapter 9: Transposes, Symmetry (A = LDLᵀ), and Row Exchanges (PA = LU)

9.1 Transpose Properties

The transpose A^T reflects the matrix across its main diagonal: $(A^T)_{ij} = A_{ji}$. The shoe-sock reversal property holds for transposition:

$$(AB)^T = B^T A^T \quad \text{and} \quad (A^{-1})^T = (A^T)^{-1}$$

9.2 Symmetric Matrices and A = LDLᵀ

A matrix is **symmetric** if $A^T = A$. When a symmetric matrix undergoes LDU factorization, symmetry forces U' to equal L^T:

A = L D L^T \left(\text{where } L \text{ is unit lower triangular, and } D = \text{diag}(d_1, \dots, d_n) \right)

Engineering Impact: You only need to compute and store L and D! The upper triangular part is automatically known, cutting computation and memory usage in half.

9.3 Factorization with Row Exchanges: PA = LU

When zero appears in a pivot position, or when partial pivoting is used for numerical accuracy, rows must be exchanged. All required row exchanges can be recorded into a single permutation matrix P:

▶ **PA = LU Factorization Theorem**

For **every invertible matrix A**, there exists a permutation matrix P such that:

$$\mathbf{P\,A = L\,U}$$

where P records all row exchanges, L is unit lower triangular with $|l_{ij}| \leq 1$, and U is upper triangular with all non-zero pivots on its diagonal.

Part V: Step-by-Step Worked Problems & Reference Appendices

Chapter 10: 5 Complete Worked Problems with Step-by-Step Tableaux

WORKED PROBLEM 1: Solving Ax = b via Gaussian Elimination & Back-Substitution

Rigorous Step-by-Step Solution

1. PROBLEM STATEMENT

Solve the following 3×3 system using complete Gaussian elimination and back-substitution. Verify both the Row and Column pictures.

2. GIVEN PARAMETERS & MATRIX FORMULATION

System:

$$\begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ 9 \end{bmatrix}$$

3. SYMBOLIC ALGEBRAIC DERIVATION

Form the augmented matrix $[A \mid \mathbf{b}]$. Identify pivot 1 ($d_1 = 2$). Compute multipliers $l_{21} = 4/2 = 2$ and $l_{31} = -2/2 = -1$. Eliminate entries in column 1. Identify pivot 2, compute l_{32}, eliminate column 2, and back-substitute.

4. NUMERICAL EXECUTION & TABLEAU

Step 1: Augmented Matrix:

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 4 & -6 & 0 & -2 \\ -2 & 7 & 2 & 9 \end{array}\right]$$

Step 2: Clear Column 1: $R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 - (-1)R_1$:

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 0 & -8 & -2 & -12 \\ 0 & 8 & 3 & 14 \end{array}\right]$$

Step 3: Clear Column 2: Pivot 2 is $d_2 = -8$. Multiplier $l_{32} = 8/(-8) = -1$. $R_3 \leftarrow R_3 - (-1)R_2$:

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 0 & -8 & -2 & -12 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

Step 4: Back-Substitution:

- From Row 3: $1 \cdot x_3 = 2$ &implies; $\mathbf{x_3 = 2}$
- From Row 2: $-8\,x_2 - 2(2) = -12$ &implies; $-8\,x_2 = -8$ &implies; $\mathbf{x_2 = 1}$
- From Row 1: $2\,x_1 + 1(1) + 1(2) = 5$ &implies; $2\,x_1 = 2$ &implies; $\mathbf{x_1 = 1}$

✓ **Verification & Sanity Check:** Column combination check: $1\begin{bmatrix} 2 \\ 4 \\ -2 \end{bmatrix} + 1\begin{bmatrix} 1 \\ -6 \\ 7 \end{bmatrix} + 2\begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 2+1+2 \\ 4-6+0 \\ -2+7+4 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ 9 \end{bmatrix} =$ **b**. Exact match!

WORKED PROBLEM 2: A = LU and A = LDU Factorization with Multiplier Synthesis

Rigorous Step-by-Step Solution

1. PROBLEM STATEMENT

Compute the complete A = LU and A = LDU factorizations for matrix A from Problem 1. Verify the reconstruction LU = A.

2. GIVEN PARAMETERS & MATRIX FORMULATION

Matrix: A =

$$\begin{vmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{vmatrix}$$

3. SYMBOLIC ALGEBRAIC DERIVATION

Record multipliers from forward elimination: $l_{21} = 2$, $l_{31} = -1$, $l_{32} = -1$. Form L by placing multipliers in lower subdiagonals and 1s on diagonal. Extract pivots $D = \text{diag}(2, -8, 1)$ to produce U' with 1s on diagonal.

4. NUMERICAL EXECUTION & TABLEAU

Multipliers extracted during elimination: $l_{21} = 2$, $l_{31} = -1$, $l_{32} = -1$.

Unit Lower Triangular Matrix L:

$$\begin{vmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -1 & 1 \end{vmatrix}$$

Upper Triangular Matrix U: $\begin{vmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{vmatrix}$

LDU Decomposition: Factoring pivots $d_1 = 2$, $d_2 = -8$, $d_3 = 1$ from rows of U:

$$A = \begin{vmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -1 & 1 \end{vmatrix} \begin{vmatrix} 2 & 0 & 0 \\ 0 & -8 & 0 \\ 0 & 0 & 1 \end{vmatrix} \begin{vmatrix} 1 & 1/2 & 1/2 \\ 0 & 1 & 1/4 \\ 0 & 0 & 1 \end{vmatrix}$$

✓ **Verification & Sanity Check:** Reconstruction: (Row 2 of L) × (Col 2 of U) = (2)(1) + (1)(-8) + (0)(0) = 2 - 8 = -6 = a_{22}. (Row 3 of L) × (Col 3 of U) = (-1)(1) + (-1)(-2) + (1)(1) = -1 + 2 + 1 = 2 = a_{33}. All 9 entries match A identically!

WORKED PROBLEM 3: Gauss-Jordan Inversion Tableau [A | I] → [I | A^{-1}]

Rigorous Step-by-Step Solution

1. PROBLEM STATEMENT

Compute the matrix inverse A^{-1} for the 3×3 matrix below using the full Gauss-Jordan augmented elimination [A | I] → [I | A^{-1}].

2. GIVEN PARAMETERS & MATRIX FORMULATION

Matrix: A =

$$\begin{vmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 1 & 0 & 2 \end{vmatrix}$$

3. SYMBOLIC ALGEBRAIC DERIVATION

Set up $[A \mid I]$. Eliminate below pivot 1 ($R_2 - 2R_1$, $R_3 - R_1$). Eliminate below pivot 2 ($R_3 + 2R_2$). Once in upper triangular form $[U \mid E]$, perform Jordan upward elimination to zero entries above the pivots. Finally, normalize diagonal entries to 1.

4. NUMERICAL EXECUTION & TABLEAU

Step 1: Setup Augmented Tableau [A | I]:

$$\left|\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 2 & 0 & 0 & 1 \end{array}\right|$$

Step 2: Forward Elimination (Column 1): $R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 - R_1$:

$$\left|\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -2 & 1 & 0 \\ 0 & -2 & 1 & -1 & 0 & 1 \end{array}\right|$$

Step 3: Forward Elimination (Column 2): $R_3 \leftarrow R_3 + 2R_2$:

$$\left|\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -2 & 1 & 0 \\ 0 & 0 & 3 & -5 & 2 & 1 \end{array}\right|$$

Step 4: Scale Row 3 (Pivot 3 → 1): $R_3 \leftarrow \frac{1}{3} R_3$:

$$\left|\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -2 & 1 & 0 \\ 0 & 0 & 1 & -5/3 & 2/3 & 1/3 \end{array}\right|$$

Step 5: Jordan Upward Elimination: Clear column 3 above pivot 3 ($R_2 \leftarrow R_2 - R_3$, $R_1 \leftarrow R_1 - R_3$):

$$\left|\begin{array}{ccc|ccc} 1 & 2 & 0 & 8/3 & -2/3 & -1/3 \\ 0 & 1 & 0 & -1/3 & 1/3 & -1/3 \\ 0 & 0 & 1 & -5/3 & 2/3 & 1/3 \end{array}\right|$$

Step 6: Clear column 2 above pivot 2: $R_1 \leftarrow R_1 - 2R_2$:

$$\left|\begin{array}{ccc|ccc} 1 & 0 & 0 & 10/3 & -4/3 & 1/3 \\ 0 & 1 & 0 & -1/3 & 1/3 & -1/3 \\ 0 & 0 & 1 & -5/3 & 2/3 & 1/3 \end{array}\right|$$

Result: $A^{-1} = \frac{1}{3}\begin{vmatrix} 10 & -4 & 1 \\ -1 & 1 & -1 \\ -5 & 2 & 1 \end{vmatrix}$

✓ **Verification & Sanity Check:** Product check: $A\,A^{-1} = \frac{1}{3}\begin{vmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 1 & 0 & 2 \end{vmatrix}\begin{vmatrix} 10 & -4 & 1 \\ -1 & 1 & -1 \\ -5 & 2 & 1 \end{vmatrix} = \frac{1}{3}$

$\begin{vmatrix} 10-2-5 & -4+2+2 & 1-2+1 \\ 20-5-15 & -8+5+6 & 2-5+3 \\ 10-0-10 & -4+0+4 & 1-0+2 \end{vmatrix} = \frac{1}{3}\begin{vmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{vmatrix}$ = $\mathbf{I}$. Perfectly verified!

WORKED PROBLEM 4: PA = LU Factorization with Zero Pivot & Row Exchange

Rigorous Step-by-Step Solution

1. PROBLEM STATEMENT

Find the PA = LU factorization for a matrix where a zero occurs in the pivot position, necessitating a row exchange.

2. GIVEN PARAMETERS & MATRIX FORMULATION

Matrix: A =

$$\begin{vmatrix} 0 & 2 & 1 \\ 2 & 1 & 1 \\ 4 & 1 & 3 \end{vmatrix}$$

3. SYMBOLIC ALGEBRAIC DERIVATION

Since $a_{11} = 0$, elimination cannot begin. Partial pivoting selects the largest entry in column 1 (entry $a_{31} = 4$ or $a_{21} = 2$). Swapping row 1 with row 2 gives non-zero pivot 2. Form permutation matrix P, perform elimination on PA, and extract L and U.

4. NUMERICAL EXECUTION & TABLEAU

Step 1: Row Exchange P_{12}: Swap row 1 and row 2:

P =

$$\begin{vmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{vmatrix} \quad \&implies; \quad PA = \begin{vmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 4 & 1 & 3 \end{vmatrix}$$

Step 2: Forward Elimination on PA: Pivot 1 is 2.
Row 2 already has a 0! Multiplier $l_{21} = 0$.
For Row 3: multiplier $l_{31} = 4/2 = 2$. $R_3 \leftarrow R_3 - 2R_1$:

$$\begin{vmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & -1 & 1 \end{vmatrix}$$

Step 3: Clear Column 2: Pivot 2 is 2. Multiplier $l_{32} = -1/2 = -0.5$. $R_3 \leftarrow R_3 - (-0.5)R_2$:

U = $$\begin{vmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1.5 \end{vmatrix}$$

Step 4: Synthesize L: $$\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & -0.5 & 1 \end{vmatrix}$$

✓ **Verification & Sanity Check:** Verify LU = PA: $$\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & -0.5 & 1 \end{vmatrix} \begin{vmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1.5 \end{vmatrix} = \begin{vmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 4 & 2\text{-}1 & 2\text{-}0.5\text{+}1.5 \end{vmatrix} =$$

$$\begin{vmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 4 & 1 & 3 \end{vmatrix}$$ = PA. Exactly holds!

WORKED PROBLEM 5: Verifying the 4 Perspectives of Matrix Multiplication

Rigorous Step-by-Step Solution

1. PROBLEM STATEMENT

Demonstrate and explicitly calculate the 4 Perspectives of Matrix Multiplication for the product AB = C.

2. GIVEN PARAMETERS & MATRIX FORMULATION

Matrices: $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 5 & 1 \\ 2 & 0 \end{bmatrix}$

3. SYMBOLIC ALGEBRAIC DERIVATION

Execute: (1) Row×Column dot products, (2) Matrix×Column combinations, (3) Row×Matrix combinations, and (4) Sum of rank-1 outer products. Verify that all 4 yield the exact same 2×2 matrix C.

4. NUMERICAL EXECUTION & TABLEAU

1. Row × Column (Inner Product):

- $c_{11} = [2, 3] \cdot [5, 2]^T = 10 + 6 = 16$
- $c_{12} = [2, 3] \cdot [1, 0]^T = 2 + 0 = 2$
- $c_{21} = [1, 4] \cdot [5, 2]^T = 5 + 8 = 13$
- $c_{22} = [1, 4] \cdot [1, 0]^T = 1 + 0 = 1$ &implies; $C = \begin{bmatrix} 16 & 2 \\ 13 & 1 \end{bmatrix}$

2. Matrix × Column (Col Combinations):

- Col 1(C) = $5\begin{bmatrix} 2 \\ 1 \end{bmatrix} + 2\begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 10+6 \\ 5+8 \end{bmatrix} = \begin{bmatrix} 16 \\ 13 \end{bmatrix}$
- Col 2(C) = $1\begin{bmatrix} 2 \\ 1 \end{bmatrix} + 0\begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ &implies; $C = \begin{bmatrix} 16 & 2 \\ 13 & 1 \end{bmatrix}$

3. Row × Matrix (Row Combinations):

- Row 1(C) = 2[5, 1] + 3[2, 0] = [10+6, 2+0] = [16, 2]
- Row 2(C) = 1[5, 1] + 4[2, 0] = [5+8, 1+0] = [13, 1] &implies; $C = \begin{bmatrix} 16 & 2 \\ 13 & 1 \end{bmatrix}$

4. Column × Row (Outer Products):

- (Col 1 of A)(Row 1 of B) = $\begin{bmatrix} 2 \\ 1 \end{bmatrix}[5, 1] = \begin{bmatrix} 10 & 2 \\ 5 & 1 \end{bmatrix}$ (Rank 1)
- (Col 2 of A)(Row 2 of B) = $\begin{bmatrix} 3 \\ 4 \end{bmatrix}[2, 0] = \begin{bmatrix} 6 & 0 \\ 8 & 0 \end{bmatrix}$ (Rank 1)
- Sum = $\begin{bmatrix} 10 & 2 \\ 5 & 1 \end{bmatrix} + \begin{bmatrix} 6 & 0 \\ 8 & 0 \end{bmatrix} = \begin{bmatrix} 16 & 2 \\ 13 & 1 \end{bmatrix}$

✓ **Verification & Sanity Check:** All 4 perspectives independently produce C = [[16, 2], [13, 1]]. Notice how the sum of rank-1 matrices decomposed C into fundamental building blocks!

Appendix A: Computational Complexity Analysis: Flop Counting (~⅓ n^3 vs n^2)

In computational mathematics, an operation (flop) historically referred to one floating-point multiplication plus one addition. How much work does it take to solve an $n \times n$ system $A\mathbf{x} = \mathbf{b}$?

A.1 Forward Elimination Flop Derivation

At step k, the submatrix remaining to be eliminated has dimension $(n - k + 1) \times (n - k + 1)$:

- There are $n - k$ rows below the pivot.
- For each row, we compute 1 multiplier: $(n - k)$ divisions.
- For each row, we multiply row k by the multiplier and subtract across $(n - k)$ entries: $(n - k)^2$ multiplications and subtractions.

Summing over all steps $k = 1$ to $n - 1$:

$$\text{Total Multiplications} = \sum_{k=1}^{n-1} (n - k)^2 = \sum_{j=1}^{n-1} j^2 = \frac{(n-1)\, n\, (2n - 1)}{6} \approx \frac{1}{3} n^3$$

Using calculus to approximate the discrete sum:

$$\sum_{j=1}^n j^2 \approx \int_0^n x^2 \, dx = \left[\frac{x^3}{3} \right]_0^n = \frac{1}{3} n^3$$

Including additions, forward elimination requires approximately **⅔ n^3 flops**.

A.2 Right-Hand Side & Back-Substitution Flop Count

Forward elimination on the right-hand side $\mathbf{b}$ (or forward solving $L\mathbf{c} = \mathbf{b}$) requires:

$$\sum_{k=1}^{n-1} (n - k) = \frac{n(n - 1)}{2} \approx \frac{1}{2} n^2 \text{ operations}$$

Back-substitution to solve $U\mathbf{x} = \mathbf{c}$ requires:

$$1 + 2 + 3 + \dots + n = \frac{n(n + 1)}{2} \approx \frac{1}{2} n^2 \text{ operations}$$

Together, the triangular solves take **n^2 operations**.

System Dimension n	Elimination Cost (⅓ n^3)	Solve Cost per b (n^2)	Ratio: Elimination / Solve
n = 10	~ 333 flops	100 flops	3.3 ×
n = 100	~ 333,333 flops	10,000 flops	33 ×
n = 1,000	~ 333,333,333 flops (0.33 Gflop)	1,000,000 flops (1 Mflop)	**333 ×**
n = 10,000	~ 3.33×10^{11} flops (333 Gflops)	10^8 flops (100 Mflops)	**3,333 ×**

Appendix B: Master Hand-Calculation Fast-Track Cheat Sheet & Matrix Rules

Operation / Concept	Standard Formula / Property	⚡ Fast-Track Shortcut / Rule of Thumb
2×2 Inverse Matrix	$\frac{1}{ad - bc}\begin{vmatrix} d & -b \\ -c & a \end{vmatrix}$	Swap diagonal entries ($a \leftrightarrow d$), negate off-diagonals ($-b$, $-c$), divide by determinant.
Elementary Inverse E_{ij}^{-1}	Flip sign of entry (i, j): $-l_{ij} \to +l_{ij}$	No inversion needed: just erase the minus sign!
L Matrix Synthesis	$L = {E_{21}}^{-1} {E_{31}}^{-1} \dots$	Drop multipliers l_{ij} directly into entry (i, j) of L below 1s on diagonal.
Product Inverse	$(AB)^{-1} = B^{-1} A^{-1}$	Shoe-sock rule: always reverse the multiplication order!
Product Transpose	$(AB)^{\mathrm{T}} = B^{\mathrm{T}} A^{\mathrm{T}}$	Order reverses! For symmetric matrices: ($A^{\mathrm{T}} A$) is always symmetric.
Permutation Inverse	$P^{-1} = P^{\mathrm{T}}$	Transposing a permutation matrix inverts it instantly.
Determinant via Pivots	$\det(A) = (-1)^r d_1 d_2 \dots d_n$	Multiply all diagonal pivots of U; flip sign once for each row exchange r.
Outer Product Sum	$AB = \sum \mathbf{a}_k \mathbf{b}_k^{\mathrm{T}}$	Each column times row is rank 1; sum them to construct the product matrix.